

Comparative Analysis: Theoretical vs. Empirical Channel Models

A comparative analysis of the evaluated channel models reveals a fundamental divergence between theoretical channel scalability and real-world spatial complexities in the context of Singular Value Thresholding (SVT) matrix completion.

Performance of Analytical Models: The mathematical frameworks—specifically the **One-Ring** and **Saleh-Valenzuela (SV)** models—exhibit highly predictable, proportional trade-offs. As environmental complexity increases, either through wider geometric angular spreads or denser multipath clustering, the Effective Rank rises monotonically. This theoretical loss of spatial correlation forces a significant increase in the required pilot overhead (from 38% to 80%) while yielding a smooth, mathematically expected degradation in (NMSE).

Performance of Empirical Models: In contrast, the measurement-based simulators. The **Weichselberger model** validates that SVT achieves optimal channel recovery (NMSE ~ 15 dB) across realistic power coupling profiles, provided the underlying spatial subspace remains structurally coherent. However, the **QuaDRiGa model** reveals a critical algorithmic limitation. Despite producing an exceptionally low Effective Rank (~ 1.34)—which theoretically indicates a highly correlated, easily estimated environment—the complex phase distortions inherent to true real-world Non-Line-of-Sight (NLOS) channels precipitate a near-complete reconstruction failure (0 dB NMSE).